

Global Enterprise Software Architecture & Systems Blueprint

DOC-SAG-009 | Comprehensive Engineering Architecture Specification v2.2

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1. EXECUTIVE SUMMARY & ARCHITECTURAL VISION

Nexora Technologies Pvt. Ltd. operates an enterprise-grade, highly available, and cloud-native software architecture designed to power our core SaaS products (including Nexora Cloud PRD-101, Nexora Secure PRD-102, Nexora Analytics PRD-103, and Nexora AI Studio PRD-104). This blueprint serves as the definitive technical standard for all engineering pods across our Bengaluru HQ, Hyderabad, Pune, Delhi, Mumbai, and Singapore locations.

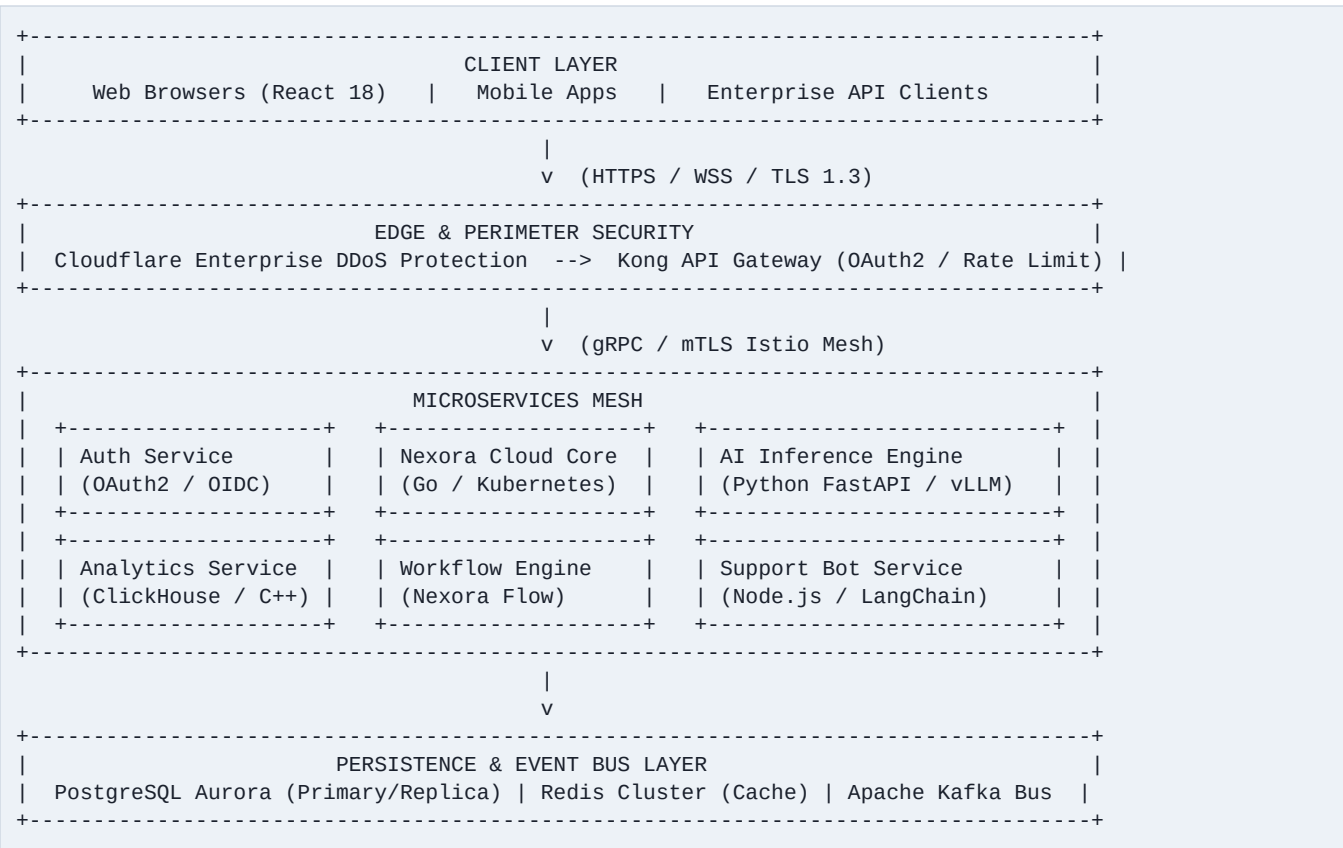
The architecture is engineered around six foundational pillars: Zero-Trust Security by Design, Microservice Decoupling, Event-Driven Streaming, Multi-Region High Availability (99.99% Uptime SLA), Continuous Observability, and Automated Failover Resilience.

1.1 Guiding Architectural Principles

- Domain-Driven Isolation:** Microservices boundaries are aligned strictly with business domain capabilities to minimize runtime dependencies.
- Zero-Trust Network Architecture:** No implicit trust exists between internal services. All internal traffic requires mutual TLS (mTLS) via Istio service mesh.
- Stateless Execution Layer:** Application pods remain completely stateless, offloading state to fault-tolerant PostgreSQL RDS Aurora and Redis clusters.
- Automated Canary Continuous Deployment:** Zero-downtime canary deployments enforced via automated GitHub Actions CI/CD workflows.

2. OVERALL SYSTEM TOPOLOGY & EDGE ARCHITECTURE

The overall system topology follows a tiered, defense-in-depth layout. All client requests originate from external web, mobile, or enterprise API callers, pass through Cloudflare Enterprise DDoS Protection, and route through our API Gateway layer into the Kubernetes microservices mesh.



2.1 Edge Security and API Gateway (Kong)

All incoming ingress traffic enters through Cloudflare Enterprise Anycast DNS with active Web Application Firewall (WAF) filtering. Approved requests reach the internal **Kong API Gateway cluster** hosted on AWS EKS across three Availability Zones in `ap-south-1` (Mumbai).

Kong executes centralized responsibilities including OAuth2/JWT token verification, API rate-limiting (token bucket algorithm), TLS 1.3 termination, and path-based routing to internal gRPC/REST service endpoints.

3. FRONTEND & MICRO-FRONTEND ARCHITECTURE

Nexora's frontend architecture standardizes on **React 18**, **TypeScript 5.2**, and **Next.js** (App Router with Server-Side Rendering). Styling is driven by Tailwind CSS and the proprietary **Nexora Design System (NDS)** component library.

3.1 Webpack 5 Module Federation

To support independent release velocity across engineering teams, the single-page application (SPA) portal utilizes **Webpack 5 Module Federation**. A host container shell dynamically loads remote micro-frontend bundles at runtime:

- **Cloud Dashboard Remote:** Developed by Core Engineering (Lead: Rohit Verma, EMP-1105). Provides control interface for Nexora Cloud (PRD-101).
- **Analytics Remote:** Developed by R&D; Pod (Lead: Dr. Ananya Sen, EMP-1490). Renders real-time telemetry and reporting for Nexora Analytics (PRD-103).
- **Support Remote:** Developed by CS Engineering Pod (Lead: Wei Chen, EMP-1201). Integrates AI chatbot and ticketing for Nexora Desk (PRD-105).

3.2 Client-Side State Management & Data Fetching

Local component state is handled via React hooks, while global application state uses **Zustand** for zero-boilerplate lightweight storage. Server-side caching, background refetching, and optimistic mutations are governed by **TanStack Query v5**.

4. BACKEND MICROSERVICES & CONTAINER ORCHESTRATION

Backend services are designed as lightweight microservices deployed on AWS Elastic Kubernetes Service (EKS). Kubernetes clusters are maintained across primary region `ap-south-1` (Mumbai) and secondary DR region `asia-southeast1` (Singapore) under the management of SysAdmin Tarun Bajaj (EMP-1455) and DevOps Lead Ali Hassan (EMP-1107).

Service Name	Tech Stack	Primary Product	Lead Engineer	Protocol
Auth Service	Go 1.22, OAuth2, Redis	Nexora Secure (PRD-102)	Tarun Bajaj (EMP-1455)	gRPC / REST
Cloud Core Service	Go 1.22, Docker SDK, AWS API	Nexora Cloud (PRD-101)	Rohit Verma (EMP-1105)	gRPC
AI Inference Engine	Python 3.11, FastAPI, vLLM, PyTorch	Nexora AI Studio (PRD-104)	Dr. Ananya Sen (EMP-1490)	gRPC / HTTPS
Analytics Query Engine	C++20, ClickHouse, Kafka	Nexora Analytics (PRD-103)	Suresh Kumar (EMP-1101)	gRPC
Workflow Engine	Java 21, Spring Boot, Postgres	Nexora Flow (PRD-106)	Sara Thomas (EMP-1106)	REST / Kafka
Support Bot Service	Node.js 20, TypeScript, LangChain	Nexora Desk (PRD-105)	Wei Chen (EMP-1201)	WebSockets / REST

4.1 Service Mesh & In-Cluster Communication

All internal pod-to-pod communication is encrypted via **Istio Service Mesh v1.20** using SPIFFE/SPIRE certificate identities. Istio sidecar proxies enforce mutual TLS (mTLS), strict circuit breaking (Max 100 concurrent requests/pod), and automatic retry policies.

5. EVENT-DRIVEN MESSAGING & DATA PIPELINES

For asynchronous workflows, decoupling, and high-volume audit logging, Nexora utilizes a multi-node **Apache Kafka** cluster stretched across three Availability Zones. Kafka topic retention and partition strategies ensure zero data loss.

Kafka Topic Name	Partition Count	Retention Period	Producer / Consumer Services
nexora.audit.events	12 Partitions	7 Years (S3 Archive)	All Services -> Security & Legal Vault (PRD-107)
nexora.telemetry.metrics	32 Partitions	30 Days	Cloud Agents -> Analytics Engine (PRD-103)
nexora.notifications.email	8 Partitions	7 Days	Workflow Engine -> SendGrid Email Dispatcher
nexora.billing.metering	12 Partitions	3 Years	SaaS Services -> Finance Payroll & Billing (DPT-04)

6. PERSISTENCE, STORAGE, & DISTRIBUTED CACHING

Nexora employs a polyglot persistence architecture tailored to workload requirements:

- **Relational Persistence:** Managed **PostgreSQL 15 (AWS RDS Aurora Multi-AZ)**. Handles transactional user accounts, subscriptions, and RBAC permissions. Primary writer in Mumbai (`ap-south-1`) with read-replicas in Singapore (`asia-southeast1`).
- **Analytical Datastore:** **ClickHouse** distributed cluster. Processes petabyte-scale telemetry metrics with columnar compression exceeding 5:1.
- **Secrets & KMS:** **Nexora Vault (PRD-107)** integrated with AWS KMS hardware security modules for dynamic DB password rotation every 24 hours.
- **Distributed In-Memory Cache:** Multi-node **Redis Cluster v7.0** managing user session tokens, rate-limiting counters, and L2 query caching with a sustained hit ratio > 92%.

7. IDENTITY, SECURITY, & AUTHORIZATION FRAMEWORK

In alignment with the Zero-Trust Network Rollout (PRJ-2002) led by Head of IT Amit Patel (EMP-1015) and CISO Karthik Natarajan (EMP-1072), all system access follows strict Identity and Access Management (IAM) controls.

7.1 Authentication & Token Lifecycle

Authentication utilizes OAuth 2.0 and OpenID Connect (OIDC) backed by Okta Enterprise. Successful logins issue short-lived, RSA-256 signed JSON Web Tokens (JWT) valid for 15 minutes, coupled with long-lived encrypted refresh tokens stored in HTTP-only secure cookies.

7.2 Authorization Engine (OPA / RBAC)

Fine-grained authorization uses **Open Policy Agent (OPA)** sidecars. Rego policies evaluate user identity, group membership, tenant ID, resource labels, and IP origin in under 2 milliseconds.

8. OBSERVABILITY, MONITORING, & INCIDENT RESPONSE

Comprehensive observability is powered by the three pillars of telemetry: Metrics, Logs, and Traces.

- **Metrics:** Prometheus scraping metrics from `/metrics` endpoints across all pods every 15 seconds, visualized via Grafana dashboards.
- **Log Aggregation:** FluentBit log shippers sending structured JSON logs to Elasticsearch (ELK Stack) with 30-day hot retention.
- **Distributed Tracing:** OpenTelemetry SDKs instrumented across Go, Python, Java, and Node services sending trace context to Jaeger.

8.1 On-Call Escalation Rota

PagerDuty manages automated on-call rotations. P0/P1 incidents trigger SMS and phone calls to Primary On-Call Engineer Rohit Verma (EMP-1105) and Secondary On-Call Sara Thomas (EMP-1106), escalating to Head of Engineering Priya Sharma (EMP-1008) if unacknowledged within 10 minutes.

9. DISASTER RECOVERY & HIGH AVAILABILITY SLAs

Nexora guarantees high availability for core enterprise production systems:

Target Metric	SLA Target Guarantee	Technical Recovery Mechanism
Monthly Platform Uptime	99.99% Availability	Multi-AZ AWS EKS deployment + Route 53 global traffic routing
Recovery Point Objective (RPO)	< 5 Minutes Data Loss	Continuous Aurora PostgreSQL cross-region WAL streaming to Singapore
Recovery Time Objective (RTO)	< 30 Minutes Downtime	Automated Terraform DR failover stack execution in GCP Singapore

10. ARCHITECTURAL GOVERNANCE & VERSIONING

The **Architecture Review Board (ARB)** convenes bi-weekly under the leadership of Principal Architect Suresh Kumar (EMP-1101) and Head of HR Rahul Desai (for staffing impact). Any proposal introducing new databases, languages, or external SaaS dependencies requires a formal Architecture Decision Record (ADR) approval.